



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Computer Studies

ITEC 204 – Data Structures and Algorithms

Name: Johnson Brix V. Santos	Date: September 24, 2026
Section: BSIT-2A	Instructor: Maxyne Nuela Ignacio

- I. **Objective:** Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

Answer:

- To apply the concepts of linear data structures in a practical IT-related application.
- To implement fundamental data structure operations, including insertion, traversal, searching, deletion, and counting
- To design a menu-driven Python application that simulates incident ticket management for automation bots.

- II. **Program Operations:** Briefly explain how your program performs each operation. (Insert Specific Code Line and Your Explanation)

Add Incident:

- Specific Code Line:

```
tickets.append(ticket)
```

Explanation:

The function prompts the user for the incident ID, bot name, and description, stores them in a dictionary object named ticket, and appends this dictionary to the end of the tickets list using the dynamic array method .append()

Display Incidents:

- Specific Code Line:

```
for ticket in tickets:  
    print(f"\nIncident ID: {ticket['incident_id']}")
```

Explanation: The function checks if the tickets list contains any items; if non-empty, it sequentially iterates through every dictionary element in the list and prints its formatted values.



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Search Incident:

- Specific Code Line

```
if ticket["incident_id"] == incident_id:
```

Explanation:

It accepts an incident ID from user input and performs a linear search across the tickets list, evaluating each record's "incident_id" key until a match is found and its complete details are displayed.

Remove Resolved Incident:

- Specific Code Line

```
tickets.remove(ticket)
```

Explanation: The function locates the target ticket by matching its "incident_id" via sequential traversal and removes that dictionary object directly from the list using tickets.remove(), dynamically updating the array size.

Count Active Incidents:

- Specific Line Code:

```
print(f"Total active tickets: {len(tickets)}")
```

Explanation:

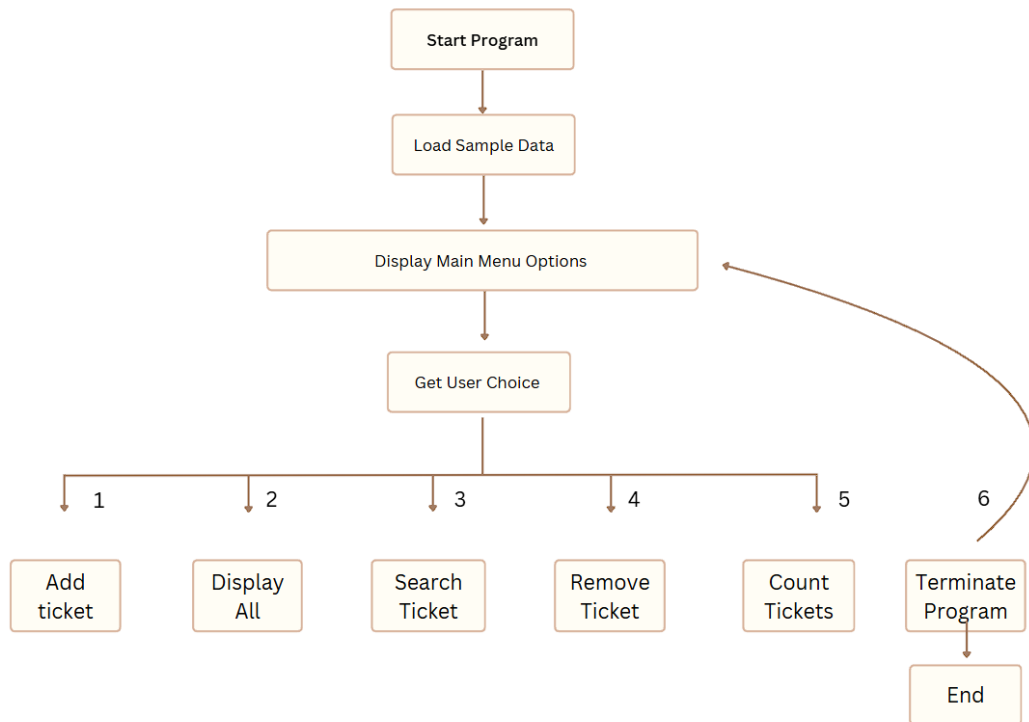
It determines the total number of active incident records currently stored in memory by passing the tickets list to Python's built-in len() function



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III. **Flowchart:** Create a flowchart showing the overall process and logic of your program.



IV. **Conclusion:** Briefly explain what you learned from the activity and how the linear data structure was applied.

Through this activity, I learned how linear data structures specifically dynamic arrays/lists containing key-value dictionaries are implemented to solve real-world system operational challenges. The activity demonstrated how linear memory structures allow efficient element insertion, sequential traversal, linear searching, and element deletion, forming the basis for effective data management in software applications.